

not gate

Truth tables :

a	not a
0	1
1	0

a	a	a nand a
0	0	1
1	1	0

So : $\text{not}(a) = a \text{ nand } a$

And in hdl : `Nand(a=in, b=in, out=out);`

and gate

Since $a \text{ nand } b = \text{not}(a \text{ and } b)$, $a \text{ and } b = \text{not}(a \text{ nand } b)$. So $a \text{ and } b = (a \text{ nand } b) \text{ nand } (a \text{ nand } b)$ based on the not gate.

In hdl, the result of $a \text{ nand } b$ is stored into $n1$ and then we compute $n1 \text{ nand } n1$:

```
Nand(a=a, b=b, out=n1);  
Nand(a=n1, b=n1, out=out);
```

or gate

Truth tables :

a	b	a or b	not(a)	not(b)	not(a) and not(b)	not(not(a) and not(b))
0	0	0	1	1	1	0
0	1	1	1	0	0	1
1	0	1	0	1	0	1
1	1	1	0	0	0	1

So : $a \text{ or } b = \text{not}(\text{not}(a) \text{ and } \text{not}(b)) = \text{not}(a) \text{ nand } \text{not}(b)$

In hdl :

```
Nand(a=a, b=a, out=nota);  
Nand(a=b, b=b, out=notb);  
Nand(a=nota, b=notb, out=out);
```

xor gate

to be continued...